

Projects in AI and Machine Learning HW 6 Part 2

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Abstract

Diffusion models have recently become one of the most powerful approaches to generative modeling, more specifically in the domain of image synthesis. Diffusion models generate data by learning to reverse a gradual noise process, which differs from GANs and VAEs. This paper explores the core principles behind diffusion models, including their forward and reverse processes, training objectives, and architectural design. A small experiment is included to analyze how the number of denoising steps affects output quality and computational efficiency. The paper concludes by discussing key limitations, including slow sampling and high computational costs, and proposes an adaptive sampling approach as a potential improvement.

1 Introduction

Generative modeling is an important problem in artificial intelligence, with applications running from image synthesis and data augmentation to scientific simulation. GANs achieve good results but often suffer from training instability, sensitivity to hyperparameters, and issues such as mode collapse. Diffusion models address these challenges by reframing the generation process as an iterative denoising problem.

In Denoising Diffusion Probabilistic Models (DDPM) by Jonathan Ho et al. [1], diffusion models demonstrate the ability to generate high-quality and diverse samples with stable training dynamics. This paper focuses on explaining the core mechanisms behind diffusion models and evaluating their behavior through a small experimental analysis.

2 Background and Related Work

Diffusion models are inspired by principles from thermodynamics and stochastic processes. The main idea is to define a forward process that gradually corrupts data with noise and then learn a reverse process that reconstructs the original data from noise.

The foundational work by Jonathan Ho et al. [1] introduced the DDPM framework, which formalizes this idea using a sequence of Gaussian noise steps. This approach simplified the generative modeling problem by breaking it into a series of small denoising tasks.

Jiaming Song et al. [2] proposed Denoising Diffusion Implicit Models (DDIM), which improved the efficiency of diffusion models by reducing the number of sampling steps and enabling deterministic generation. These advancements made diffusion models more practical for real-world applications.

3 Methodology

Diffusion models consist of two main processes: a forward noising process and a reverse denoising process.

3.1 Forward Process

The forward process gradually adds Gaussian noise to a data sample x_0 over a series of timesteps. At each step, a small amount of noise is introduced, which eventually transforms the data into nearly pure noise. This process is mathematically defined as a Markov chain, where each step depends only on the previous one. The key property of this process is that it is fixed and does not require learning. It serves to define a structured way of corrupting the data.

3.2 Reverse Process

The reverse process is where a neural network is trained to reverse the effects of noise by predicting either the noise itself or the original data. This is implemented using a U-Net architecture, which is well-suited for image data because of its ability to capture both local and global features.

The model takes a noisy image and a timestep as input and outputs a prediction of the noise that was added. By iteratively applying this denoising process, the model is able to generate realistic images starting from random noise.

3.3 Training Objective

The training objective of diffusion models is relatively simple and involves minimizing the difference between the predicted noise and the actual noise added during the forward process. This is expressed as a mean-squared error loss.

This formulation is powerful because it transforms the complex problem of data generation into a supervised learning task, where the model learns to predict noise at different levels of corruption.

4 Hands-On Experiment: Effect of Denoising Steps

To better understand how diffusion models operate in practice, we analyze how the number of denoising steps affects the quality and efficiency of generated outputs. In this experiment, different configurations are compared using a pretrained Stable Diffusion model with a fixed prompt, fixed seed, and identical model checkpoint to ensure controlled comparison. The number of inference steps is varied while keeping all other settings constant. The observed outputs follow the same trend shown in the table, where increased steps lead to improved image quality at the cost of higher computational time.

Link to experiment:

<https://colab.research.google.com/drive/1GkDam3fkVOPpAJGglWnbUqn63iD4miPl?usp=sharing>

Results:

Number of Steps	Output Quality	Sharpness	Sampling Time	Artifacts
50	Good	Moderately sharp	2.32	Slight/minor
200	Very good	Sharp	8.78	Very few
999	Very good	Very sharp	43.15	Minimal

Table 1: Effect of increasing denoising steps on diffusion model output quality and sampling efficiency.

Analysis:

The results show diminishing returns in output quality as the number of denoising steps increases. At 50 steps, the model already produces a visually realistic image, demonstrating that most global structure is formed early in the denoising process. Increasing the number of steps to 200 has modest improvements in fine detail and sharpness, but the differences are subtle. At ≈ 1000 steps, the image shows minimal perceptual improvement compared to 200 steps, while significantly increasing computation time.

The primary effect of increasing denoising steps is improved refinement of small details rather than major changes in image quality. This highlights that diffusion models reach a near-optimal output within relatively few steps, motivating efficient sampling methods such as DDIM and other accelerated schedulers.

5 Discussion

Diffusion models provide a different approach to generative modeling compared to GANs and VAEs. They are able to achieve greater stability during training and produce high-quality outputs by learning iteratively to denoise data.

However, the iterative process also introduces a high computational cost during inference, as many steps are required to generate a single sample. The strong performance of diffusion models has made them a dominant approach in modern generative AI, despite this drawback.

This experiment shows that the quality of the output is directly related to the number of refinement steps applied during inference. This insight has led to the development of more efficient sampling methods such as DDIM.

6 Limitations and Challenges

One major limitation of diffusion models is their slow sampling process. Generation requires many iterative denoising steps, so inference can be significantly slower compared to GAN-based methods, which generate images in a single forward pass.

Another challenge is the high computational cost associated with both training and inference. Diffusion models often require large datasets and substantial GPU resources, which can limit accessibility and scalability.

In conditional settings such as text-to-image generation, diffusion models can be sensitive to input prompts, which makes it difficult to achieve precise control over generated inputs.

7 Proposed Extension

An extension to diffusion models is the use of adaptive step sampling. Instead of using a fixed number of denoising steps, the model could dynamically adjust the number of steps based on the complexity of the image being generated.

The model could allocate more steps to regions where the noise prediction error is high and fewer steps where the image has already stabilized. This could be improved by reducing unnecessary computation while maintaining high output quality.

A potential experiment would involve comparing a fixed-step diffusion model with an adaptive-step version. Metrics such as sampling time and output quality could be used to evaluate performance.

8 Conclusion

Diffusion models represent a significant advancement in generative modeling by reframing the generation process as an iterative denoising task. This approach leads to stable training and high-quality outputs, which makes diffusion models one of the most effective generative methods available today.

There are still challenges, such as slow sampling and high computational requirements, despite the strengths of the model. Future research, including adaptive sampling strategies and improved architectures, may help address these limitations and further enhance the practicality of diffusion models.

References

- [1] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. Advances in Neural Information Processing Systems. <https://arxiv.org/abs/2006.11239>
- [2] Song, J., Meng, C., & Ermon, S. (2020). Denoising diffusion implicit models. International Conference on Learning Representations. <https://arxiv.org/abs/2010.02502>